

# Sheet C-101 - General Plan and Control Notes

| Control item          | Revision F value              | Superseded / conflict value      | Instruction                                                             |
|-----------------------|-------------------------------|----------------------------------|-------------------------------------------------------------------------|
| Plan/profile source   | CC-RD-REV-F-2026-06           | 2019 CAD export                  | Use Revision F geometry for roadway, ditch, culvert and bus-pad layout. |
| Project datum         | CCVD 2026                     | Field-book FB-03 temporary basis | Apply signed +0.18 m survey correction to raw survey elevations.        |
| Culvert CUL-01        | 900 mm RCP at Sta. 5+80       | 750 mm CMP from maintenance form | Use 900 mm RCP from this plan set; keep outlet debris inspection open.  |
| Roadway cross-section | 6.0 m lanes + 1.2 m shoulders | 2019 sketch with 5.5 m lanes     | Use Revision F cross-section for tie-ins.                               |

Plan diagram note: Industrial access road runs east-west from Station 0+00 to Station 9+80. CUL-01 crosses the north roadside ditch at Station 5+80. Bus stop pad is near Station 2+15. Driveway D-3 is near Station 8+70 and requires grade-break checking.

Sheet C-201 - Roadway Profile and Pavement Design Inputs

| Station | Existing CL elev. m | Proposed B profile elev. m | Roadside ditch invert m | Design note                                              |
|---------|---------------------|----------------------------|-------------------------|----------------------------------------------------------|
| 0+00    | 112.180             | 112.230                    | 111.220                 | Tie to existing pavement                                 |
| 2+15    | 111.335             | 111.360                    | 110.540                 | Bus pad ADA check uses survey edge points                |
| 4+80    | 110.962             | 111.020                    | 109.620                 | Ditch capacity controls drainage option                  |
| 5+80    | 110.845             | 110.910                    | 108.902                 | CUL-01 inlet invert from corrected survey                |
| 7+20    | 110.696             | 110.820                    | 109.110                 | Utility potholing open item                              |
| 8+70    | 110.224             | 110.400                    | 109.060                 | D-3 tie-in grade adjustment needed if profile not raised |
| 9+80    | 110.050             | 110.220                    | 108.980                 | End of design limits                                     |

| Layer coefficient / material   | Value         | Source note                                                                             |
|--------------------------------|---------------|-----------------------------------------------------------------------------------------|
| a1 - HMA                       | 0.44 per inch | Project pavement-design convention based on public flexible-pavement method terminology |
| a2 - crushed aggregate base    | 0.14 per inch | Use with drainage coefficient m2                                                        |
| m2 - base drainage coefficient | 0.85          | Approved by design lead for Cedar Creek soil conditions                                 |
| Design CBR                     | 3.0 percent   | Signed geotechnical correction GEO-LAB-2026-06                                          |
| Required SN                    | 3.20          | Owner decision DEC-STR-01                                                               |

Sheet C-301 - Roadside Ditch and Culvert CUL-01

| Item                       | Revision F value      | Conflict / missing fact                              | Design instruction                                                        |
|----------------------------|-----------------------|------------------------------------------------------|---------------------------------------------------------------------------|
| Ditch catchment area       | 4.60 ha               | 2019 GIS includes undeveloped west parcel at 5.05 ha | Use 4.60 ha because west parcel drains to county channel after 2025 berm. |
| Runoff coefficient         | 0.72                  | Maintenance note uses 0.60 for grassed area only     | Use composite value from active impervious and gravel areas.              |
| 10-year rainfall intensity | 96 mm/hr              | Old 2-year maintenance storm note gives 58 mm/hr     | Use 10-year design basis from signed constraints.                         |
| Corrected design flow      | 0.884 m3/s            | No conflict                                          | Derived with Q = 0.00278 CIA.                                             |
| CUL-01 barrel              | 900 mm RCP            | Maintenance form says 750 mm CMP                     | Use Rev F barrel; inspect debris as open item.                            |
| CUL-01 length              | 28.4 m                | No conflict                                          | Use for culvert hydraulic check.                                          |
| CUL-01 inlet/outlet invert | 108.902 m / 108.722 m | No conflict after datum correction                   | Use for headwater and tailwater calculations.                             |
| Entrance condition         | Square-edge headwall  | No wingwall detail issued                            | Use square-edge entrance treatment unless updated detail is issued.       |

Alternative screening note: Alternative B upsizes CUL-01 to 1200 mm RCP and reshapes the north ditch. Alternative A retains the existing culvert and is expected to fail the HW/D screen. Alternative C diverts flow through Parcel 12 and requires a signed easement. Alternative D replaces the culvert with a concrete box but exceeds budget and closure limits.

Sheet C-401 - Bus Stop Pad, Driveways and Conflict Register

| Element                  | Revision F / corrected survey value            | Requirement / issue                         | Instruction                                                          |
|--------------------------|------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------|
| Bus pad cross-slope      | BUS-N 111.344 m; BUS-S 111.326 m over 1.80 m   | Accessible route cross-slope <= 2.0 percent | Calculate from corrected survey edge elevations.                     |
| Driveway D-2 tie-in      | 6.48 percent grade break                       | Maximum local tie-in grade 8.0 percent      | Pass.                                                                |
| Driveway D-3 tie-in      | 8.98 percent grade break if profile not raised | Maximum local tie-in grade 8.0 percent      | Fails under Alternative A; Alternative B raises profile to mitigate. |
| Utility potholing        | Incomplete at Sta. 6+40 to 7+10                | Unknown water and telecom depth             | Open item; not a 60% blocker for B if listed.                        |
| Parcel 12 drainage swale | Requires easement                              | Signed easement not supplied                | Blocks Alternative C.                                                |

| Conflict ID | Superseded / uncertain claim                      | Controlling source                  | Resolution                                   |
|-------------|---------------------------------------------------|-------------------------------------|----------------------------------------------|
| PLN-01      | 2019 CAD says 750 mm culvert                      | Sheet C-301 Revision F              | Use 900 mm RCP.                              |
| PLN-02      | Old D-3 profile ties without raising roadway      | Sheet C-201 Revision F + D-3 survey | Alternative B must include grade adjustment. |
| PLN-03      | Phone-level ADA note says 2.7 percent cross-slope | Corrected survey BUS-N/BUS-S        | Use survey calculation, not phone note.      |
| PLN-04      | Parcel 12 swale drawn as future option            | Owner decision DEC-ROW-01           | Cannot assume unsigned easement.             |